

Production 6 :

La supervision reseau et systeme avec Nagios Core sous Ubuntu Server

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Module : E5 - Support et mise a disposition
de services informatiques

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1. Introduction

Nagios est une application permettant la surveillance système et réseau. Elle surveille les hôtes et services spécifiés, alertant lorsque les systèmes dysfonctionnent et quand ils passent en fonctionnement normal. C'est un logiciel libre sous licence GPL.

L'interface web permet d'avoir une vue d'ensemble du système d'information et des possibles anomalies.

2. Topologie et Configuration reseau

Machine Ubuntu 20.04 LTS avec deux interfaces reseau.

Hostname : nagios

Carte 1 : Bridged / NAT

Carte 2 : IP statique en segment LAN

Les etapes suivantes ont ete realisees :

```
sudo apt install ifupdown
ip a
sudo nano /etc/network/interfaces
sudo nano /etc/hostname
sudo nano /etc/hosts
ping <IP>
```

```
root@nagios:/home/nagioshenrio# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:62:85:67 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.146.145/24 metric 100 brd 192.168.146.255 scope global dynamic ens33
        valid_lft 1409sec preferred_lft 1409sec
    inet6 fe80::20c:29ff:fe62:8567/64 scope link
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 00:0c:29:62:85:71 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
root@nagios:/home/nagioshenrio#
```

Figure 1 : Verification des interfaces avec ip a apres installation d'ifupdown

```

GNU nano 6.2 /etc/network/interfaces *
# interfaces(5) file used by ifup(8) and ifdown(8)
# Include files from /etc/network/interfaces.d:
source /etc/network/interfaces.d/*
auto lo
iface lo inet loopback

auto ens33
iface ens33 inet dhcp

auto ens34
iface ens34 inet static
address 172.20.0.31/24

```

Figure 2 : Configuration du fichier /etc/network/interfaces

```

root@nagios:/home/nagioshenrio# ^C
root@nagios:/home/nagioshenrio# ^[[200~nano /etc/network/interfaces
nano: command not found
root@nagios:/home/nagioshenrio# nano /etc/network/interfaces
root@nagios:/home/nagioshenrio# service networking restart
ifup ens34
ifup: interface ens34 already configured
root@nagios:/home/nagioshenrio# ip a show ens34
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:62:85:71 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 172.20.0.31/24 brd 172.20.0.255 scope global ens34
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe62:8571/64 scope link
        valid_lft forever preferred_lft forever

```

Figure 3 : Configuration de l'interface ens34 en IP statique

```
root@nagios:/home/nagioshenrio# ping -c 2 1.1.1.1
ping -c 2 google.fr
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=128 time=8.06 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=128 time=6.99 ms

--- 1.1.1.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 6.991/7.523/8.056/0.532 ms
PING google.fr (172.217.22.67) 56(84) bytes of data.
64 bytes from fra15s17-in-f3.1e100.net (172.217.22.67): icmp_seq=1 ttl=128 time=
6.71 ms
64 bytes from fra15s17-in-f3.1e100.net (172.217.22.67): icmp_seq=2 ttl=128 time=
5.79 ms

--- google.fr ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 5.792/6.249/6.707/0.457 ms
root@nagios:/home/nagioshenrio#
```

Figure 4 : Validation de la connectivite avec ping

3. Prerequis

Installation des packages prerequis necessaires a la compilation et au fonctionnement de Nagios Core :

```
sudo apt install -y autoconf gcc libc6 make wget unzip  
sudo apt install -y apache2 php libapache2-mod-php  
sudo apt install -y libgd-dev libmcrypt-dev libssl-dev  
sudo apt install -y dc snmp libnet-snmp-perl gettext
```

4. Telechargement et compilation de Nagios Core

Telechargement de l'archive Nagios Core depuis le depot GitHub officiel, decompression, puis compilation avec `./configure` et `make all`.

```
cd /tmp
wget https://github.com/NagiosEnterprises/nagioscore/releases/...
tar xzf nagioscore-nagios-4.4.6.tar.gz
cd nagioscore-nagios-4.4.6
./configure --with-httpd-conf=/etc/apache2/sites-enabled
make all
```

```
root@nagios:/home/nagioshenrio# cd /tmp
wget -O nagioscore.tar.gz https://github.com/NagiosEnterprises/nagioscore/archive/nagios-4.4.8.tar.gz
tar xvzf nagioscore.tar.gz
--2026-03-17 12:51:18-- https://github.com/NagiosEnterprises/nagioscore/archive/nagios-4.4.8.tar.gz
Resolving github.com (github.com)... 140.82.121.4
Connecting to github.com (github.com)|140.82.121.4|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://codeload.github.com/NagiosEnterprises/nagioscore/tar.gz/refs/tags/nagios-4.4.8 [following]
--2026-03-17 12:51:18-- https://codeload.github.com/NagiosEnterprises/nagioscore/tar.gz/refs/tags/nagios-4.4.8
Resolving codeload.github.com (codeload.github.com)... 140.82.121.9
Connecting to codeload.github.com (codeload.github.com)|140.82.121.9|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [application/x-gzip]
Saving to: 'nagioscore.tar.gz'

nagioscore.tar.gz      [          <=>          ] 10,81M  4,06MB/s   in 2,7s

2026-03-17 12:51:21 (4,06 MB/s) - 'nagioscore.tar.gz' saved [11339506]
```

Figure 5 : Telechargement de Nagios Core depuis GitHub

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@nagios:~# cd /tmp/nagioscore-nagios-4.4.8/
./configure --with-httpd-conf=/etc/apache2/sites-enabled
checking for a BSD-compatible install... /usr/bin/install -c
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether we are using the GNU C compiler... yes
checking whether gcc accepts -g... yes
checking for gcc option to accept ISO C89... none needed
checking whether make sets $(MAKE)... yes
checking whether ln -s works... yes
checking for strip... /usr/bin/strip
checking how to run the C preprocessor... gcc -E
checking for grep that handles long lines and -e... /usr/bin/grep
checking for egrep... /usr/bin/grep -E
checking for ANSI C header files... yes
checking whether time.h and sys/time.h may both be included... yes
checking for sys/wait.h that is POSIX.1 compatible... yes
checking for sys/types.h... yes
checking for sys/stat.h... yes
checking for stdlib.h... yes
checking for string.h... yes
checking for memory.h... yes
checking for strings.h... yes
checking for inttypes.h... yes
checking for stdint.h... yes
checking forunistd.h... yes
checking arpa/inet.h usability... yes
```

Figure 6 : Execution de ./configure pour Nagios Core


```

root@nagios:/tmp/nagioscore-nagios-4.4.8# make all
cd ./base && make
make[1]: Entering directory '/tmp/nagioscore-nagios-4.4.8/base'
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o nagios.o nagios.c
nagios.c: In function 'main':
nagios.c:611:25: warning: ignoring return value of 'asprintf' declared with attribute 'warn_unused_result' [-Wunused-result]
  611 |         asprintf(&mac->x[MACRO_PROCESSTARTTIME], "%llu"
      |         ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
nagios.c:841:25: warning: ignoring return value of 'asprintf' declared with attribute 'warn_unused_result' [-Wunused-result]
  841 |         asprintf(&mac->x[MACRO_EVENTSTARTTIME], "%llu",
      |         ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
nagios.c: In function 'nagios_core_worker':
nagios.c:176:17: warning: ignoring return value of 'read' declared with attribute 'warn_unused_result' [-Wunused-result]
  176 |         read(sd, response + 3, sizeof(response) - 4);
      |         ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
nagios.c: In function 'test_path_access':
nagios.c:122:17: warning: ignoring return value of 'asprintf' declared with attribute 'warn_unused_result' [-Wunused-result]
  122 |         asprintf(&path, "%s/%s", p, program);
      |         ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o broker.o broker.c
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o nebmods.o nebmods.c
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o ../common/shared.o ../common/shared.c
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o query-handler.o query-handler.c
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o workers.o workers.c
workers.c: In function 'handle_worker_result':
workers.c:801:25: warning: ignoring return value of 'asprintf' declared with attribute 'warn_unused_result' [-Wunused-result]
  801 |         asprintf(&error_reason, "timed out after %.2fs",
      |         ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
tv_delta_f(&wpres.start, &wpres.stop));

```

Figure 7 : Compilation avec make all

5. Installation

Creation de l'utilisateur et du groupe nagios, puis installation des binaires, du daemon, du mode commande et des fichiers de configuration d'exemple.

```
sudo make install-groups-users
sudo usermod -aG nagios www-data
sudo make install
sudo make install-daemoninit
sudo make install-commandmode
sudo make install-config
```

```
*** Init script installed ***

/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/var/rw
chmod g+s /usr/local/nagios/var/rw

*** External command directory configured ***

/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc/objects
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/nagios.cfg /usr/
local/nagios/etc/nagios.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/cgi.cfg /usr/loc
al/nagios/etc/cgi.cfg
/usr/bin/install -c -b -m 660 -o nagios -g nagios sample-config/resource.cfg /us
r/local/nagios/etc/resource.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/
templates.cfg /usr/local/nagios/etc/objects/templates.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/
commands.cfg /usr/local/nagios/etc/objects/commands.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/
contacts.cfg /usr/local/nagios/etc/objects/contacts.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/
timeperiods.cfg /usr/local/nagios/etc/objects/timeperiods.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/
localhost.cfg /usr/local/nagios/etc/objects/localhost.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/
windows.cfg /usr/local/nagios/etc/objects/windows.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/
printer.cfg /usr/local/nagios/etc/objects/printer.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/
switch.cfg /usr/local/nagios/etc/objects/switch.cfg

*** Config files installed ***

Remember, these are *SAMPLE* config files. You'll need to read
the documentation for more information on how to actually define
services, hosts, etc. to fit your particular needs.

root@nagios:/tmp/nagioscore-nagios-4.4.8#
```

Figure 8 : Creation de l'utilisateur et du groupe nagios

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@nagios:/home/nagioshenrio# cd /tmp/nagioscore-nagios-4.4.8/
make install-webconf
a2enmod rewrite
systemctl restart apache2
a2enmod cgi
systemctl restart apache2
/usr/bin/install -c -m 644 sample-config/httpd.conf /etc/apache2/sites-enabled/nagios.conf
if [ 0 -eq 1 ]; then \
    ln -s /etc/apache2/sites-enabled/nagios.conf /etc/apache2/sites-enabled/nagios.conf; \
fi

*** Nagios/Apache conf file installed ***

Enabling module rewrite.
To activate the new configuration, you need to run:
    systemctl restart apache2
Enabling module cgi.
To activate the new configuration, you need to run:
    systemctl restart apache2
root@nagios:/tmp/nagioscore-nagios-4.4.8#
```

Figure 9 : Navigation dans le dossier /tmp pour l'installation

6. Configuration Apache

Installation des fichiers de configuration Apache pour Nagios, activation des modules rewrite et cgi.

```
sudo make install-webconf
sudo a2enmod rewrite
sudo a2enmod cgi
```

```
root@nagios:/tmp/nagioscore-nagios-4.4.8# make install-webconf
a2enmod rewrite
systemctl restart apache2
a2enmod cgi
systemctl restart apache2
/usr/bin/install -c -m 644 sample-config/httpd.conf /etc/apache2/sites-enabled/nagios.conf
if [ 0 -eq 1 ]; then \
    ln -s /etc/apache2/sites-enabled/nagios.conf /etc/apache2/sites-enabled/nagios.conf; \
fi

*** Nagios/Apache conf file installed ***

Module rewrite already enabled
Module cgi already enabled
root@nagios:/tmp/nagioscore-nagios-4.4.8#
```

Figure 10 : Installation de la configuration web Apache

7. Configuration du Firewall

Autorisation du trafic HTTP entrant sur le pare-feu UFW afin de permettre l'accès à l'interface web Nagios.

```
sudo ufw allow Apache
```

```
root@nagios:/tmp/nagioscore-nagios-4.4.8# ufw allow Apache
ufw reload
Rules updated
Rules updated (v6)
Firewall not enabled (skipping reload)
root@nagios:/tmp/nagioscore-nagios-4.4.8#
```

Figure 11 : Autorisation d'Apache dans le pare-feu UFW

8. Creation du compte nagiosadmin

Creation du compte utilisateur Apache pour l'accès à l'interface web Nagios. Ce compte sera utilisé pour se connecter à l'interface d'administration.

```
sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
```

```
root@nagios:/tmp/nagioscore-nagios-4.4.8# htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
root@nagios:/tmp/nagioscore-nagios-4.4.8#
```

Figure 12 : Creation du mot de passe nagiosadmin

9. Demarrage des services

Demarrage et verification du bon fonctionnement des services Apache2 et Nagios.

```
sudo systemctl start apache2
sudo systemctl status apache2
sudo systemctl start nagios
sudo systemctl status nagios
```

```
root@nagios:/tmp/nagioscore-nagios-4.4.8# systemctl restart apache2.service
systemctl status apache2.service
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-03-17 14:54:52 UTC; 51ms ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 52111 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 52120 (apache2)
    Tasks: 1 (limit: 2344)
   Memory: 3.9M
      CPU: 263ms
   CGroup: /system.slice/apache2.service
           └─52120 /usr/sbin/apache2 -k start

mars 17 14:54:52 nagios systemd[1]: Starting The Apache HTTP Server...
mars 17 14:54:52 nagios systemd[1]: Started The Apache HTTP Server.
root@nagios:/tmp/nagioscore-nagios-4.4.8#
```

Figure 13 : Demarrage et verification du service Apache2

```
root@nagios:/tmp/nagioscore-nagios-4.4.8# systemctl start nagios.service
systemctl status nagios.service
● nagios.service - Nagios Core 4.4.8
   Loaded: loaded (/lib/systemd/system/nagios.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-03-17 14:55:25 UTC; 69ms ago
     Docs: https://www.nagios.org/documentation
   Process: 52317 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cf
   Process: 52318 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
  Main PID: 52319 (nagios)
    Tasks: 1 (limit: 2344)
   Memory: 800.0K
      CPU: 67ms
   CGroup: /system.slice/nagios.service
           └─52319 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             └─52322 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
               └─52323 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
                 └─52324 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                   └─52325 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh

mars 17 14:55:25 nagios nagios[52317]: Things look okay - No serious problems were detected dur
mars 17 14:55:25 nagios systemd[1]: Started Nagios Core 4.4.8.
mars 17 14:55:25 nagios nagios[52319]: Nagios 4.4.8 starting... (PID=52319)
mars 17 14:55:25 nagios nagios[52319]: Local time is Tue Mar 17 14:55:25 UTC 2026
mars 17 14:55:25 nagios nagios[52319]: LOG VERSION: 2.0
mars 17 14:55:25 nagios nagios[52319]: qh: Socket '/usr/local/nagios/var/rw/nagios.qh' successf
mars 17 14:55:25 nagios nagios[52319]: qh: core query handler registered
mars 17 14:55:25 nagios nagios[52319]: qh: echo service query handler registered
mars 17 14:55:25 nagios nagios[52319]: qh: help for the query handler registered
mars 17 14:55:25 nagios nagios[52319]: wproc: Successfully registered manager as @wproc with qu
lines 1-27/27 (END)
```

Figure 14 : Demarrage et verification du service Nagios

10. Test de l'interface web Nagios

Acces a l'interface web Nagios via le navigateur a l'adresse `http://<IP_serveur>/nagios`. Connexion avec le compte nagiosadmin.



The screenshot displays the Nagios Core web interface. On the left is a sidebar menu with categories: General (Home, Documentation), Current Status (Tactical Overview, Map (Legacy), Hosts, Services, Host Groups, Summary, Grid, Service Groups, Summary, Grid), Problems (Services (Unhandled), Hosts (Unhandled), Network Outages), Quick Search, Reports (Availability, Trends (Legacy), Alerts, History, Summary, Histogram (Legacy), Notifications, Event Log), and System (Comments, Downtime, Process Info, Performance Info, Scheduling Queue, Configuration). The main content area features the Nagios Core logo, a status message 'Daemon running with PID 52319', the version 'Nagios® Core™ Version 4.4.8' dated 'October 04, 2022' with a 'Check for updates' link, and a blue notification box stating 'A new version of Nagios Core is available! Visit nagios.org to download Nagios 4.5.11.' Below this are four boxes: 'Get Started' with links to start monitoring, change look, extend Nagios, get support, get training, and get certified; 'Quick Links' with links to Nagios Library, Nagios Labs, Nagios Exchange, Nagios Support, Nagios.com, and Nagios.org; 'Latest News'; and 'Don't Miss...'. At the bottom, there is a copyright notice and a license disclaimer.

Nagios®

General
Home
Documentation

Current Status
Tactical Overview
Map (Legacy)
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
[input type="text"]

Reports
Availability
Trends (Legacy)
Alerts
History
Summary
Histogram (Legacy)
Notifications
Event Log

System
Comments
Downtime
Process Info
Performance Info
Scheduling Queue
Configuration

Nagios® Core™
✓ Daemon running with PID 52319
Nagios® Core™
Version 4.4.8
October 04, 2022
[Check for updates](#)

A new version of Nagios Core is available!
Visit nagios.org to download Nagios 4.5.11.

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Figure 15 : Interface web Nagios Core operationnelle

11. Installation des plugins Nagios

Installation des prerequis, telechargement et compilation des plugins nagios-plugins 2.2.1 pour activer les controles de supervision.

```
cd /tmp
wget https://github.com/nagios-plugins/.../nagios-plugins-2.2.1.tar.gz
tar xzf nagios-plugins-2.2.1.tar.gz
cd nagios-plugins-2.2.1
./configure
make
sudo make install
```

```
check_rpc check_sensors check_wave check_ifstatus check_ifoperstatus check_mailq check_file_age
utils.sh utils.pm '/usr/local/nagios/libexec'
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/tmp/nagios-plugins-release-2.2.1/plugins-scripts'
make[1]: Leaving directory '/tmp/nagios-plugins-release-2.2.1/plugins-scripts'
Making install in plugins-root
make[1]: Entering directory '/tmp/nagios-plugins-release-2.2.1/plugins-root'
2.2.1.git
make[2]: Entering directory '/tmp/nagios-plugins-release-2.2.1/plugins-root'
2.2.1.git
/usr/bin/install -c check_dhcp /usr/local/nagios/libexec/check_dhcp
chown root /usr/local/nagios/libexec/check_dhcp
chmod ug=rx,u+s /usr/local/nagios/libexec/check_dhcp
/usr/bin/install -c check_icmp /usr/local/nagios/libexec/check_icmp
chown root /usr/local/nagios/libexec/check_icmp
chmod ug=rx,u+s /usr/local/nagios/libexec/check_icmp
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/tmp/nagios-plugins-release-2.2.1/plugins-root'
make[1]: Leaving directory '/tmp/nagios-plugins-release-2.2.1/plugins-root'
Making install in po
make[1]: Entering directory '/tmp/nagios-plugins-release-2.2.1/po'
/usr/bin/mkdir -p /usr/local/nagios/share
installing fr.gmo as /usr/local/nagios/share/locale/fr/LC_MESSAGES/nagios-plugins.mo
installing de.gmo as /usr/local/nagios/share/locale/de/LC_MESSAGES/nagios-plugins.mo
if test "nagios-plugins" = "gettext-tools"; then \
  /usr/bin/mkdir -p /usr/local/nagios/share/gettext/po; \
  for file in Makefile.in.in remove-potcdate.sin Makevars.template; do \
    /usr/bin/install -c -m 644 ./.$file \
      /usr/local/nagios/share/gettext/po/$file; \
  done; \
  for file in Makevars; do \
    rm -f /usr/local/nagios/share/gettext/po/$file; \
  done; \
else \
  :; \
fi
make[1]: Leaving directory '/tmp/nagios-plugins-release-2.2.1/po'
make[1]: Entering directory '/tmp/nagios-plugins-release-2.2.1'
make[2]: Entering directory '/tmp/nagios-plugins-release-2.2.1'
make[2]: Nothing to be done for 'install-exec-am'.
make[2]: Nothing to be done for 'install-data-am'.
make[2]: Leaving directory '/tmp/nagios-plugins-release-2.2.1'
make[1]: Leaving directory '/tmp/nagios-plugins-release-2.2.1'
root@nagios:/tmp/nagios-plugins-release-2.2.1#
```

Figure 16 : Fin de l'installation des plugins Nagios

12. Activation de la supervision

Edition du fichier `nagios.cfg` pour commenter les inclusions des fichiers `windows.cfg`, `switch.cfg` et `printer.cfg`, ce qui active la supervision de ces types d'equipements.

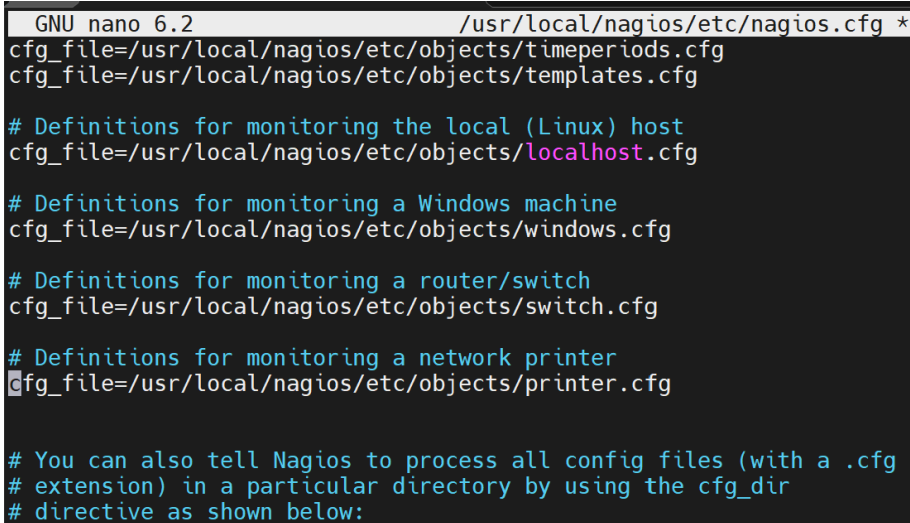
```
sudo nano /usr/local/nagios/etc/nagios.cfg
```

```
# Decommenter les lignes suivantes :
```

```
cfg_file=/usr/local/nagios/etc/objects/windows.cfg
```

```
cfg_file=/usr/local/nagios/etc/objects/switch.cfg
```

```
cfg_file=/usr/local/nagios/etc/objects/printer.cfg
```



```
GNU nano 6.2 /usr/local/nagios/etc/nagios.cfg *
cfg_file=/usr/local/nagios/etc/objects/timeperiods.cfg
cfg_file=/usr/local/nagios/etc/objects/templates.cfg

# Definitions for monitoring the local (Linux) host
cfg_file=/usr/local/nagios/etc/objects/localhost.cfg

# Definitions for monitoring a Windows machine
cfg_file=/usr/local/nagios/etc/objects/windows.cfg

# Definitions for monitoring a router/switch
cfg_file=/usr/local/nagios/etc/objects/switch.cfg

# Definitions for monitoring a network printer
cfg_file=/usr/local/nagios/etc/objects/printer.cfg

# You can also tell Nagios to process all config files (with a .cfg
# extension) in a particular directory by using the cfg_dir
# directive as shown below:
```

Figure 17 : Decommentation des fichiers de configuration dans `nagios.cfg`

13. Supervision des machines Windows

Edition du fichier windows.cfg pour ajouter les hotes DC et SDC. Modification des blocs define host et define service pour superviser les machines Windows du reseau.

```
sudo nano /usr/local/nagios/etc/objects/windows.cfg
```

```
GNU nano 6.2 /usr/local/nagios/etc/objects/windows.cfg *
#####
# WINDOWS.CFG - SAMPLE CONFIG FILE FOR MONITORING A WINDOWS MACHINE
#
#
# NOTES: This config file assumes that you are using the sample configuration
#       files that get installed with the Nagios quickstart guide.
#
#####
#####
#
# HOST DEFINITIONS
#
#####

# Define a host for the Windows machine we'll be monitoring
# Change the host_name, alias, and address to fit your situation

define host{
    use                windows-server
    host_name          DC
    alias              AD-DS
    address             172.20.0.10
}

define host{
    use                windows-server
    host_name          SDC
    alias              My secondary Windows Server
    address             172.20.0.11
}
```

Figure 18 : Modification du define host dans windows.cfg

```

define service {
    use                generic-service
    host_name          DC,SDC
    service_description Memory Usage
    check_command       check_nt!MEMUSE!-w 80 -c 90
}

# Create a service for monitoring C:\ disk usage
# Change the host_name to match the name of the host you defined above

define service {
    use                generic-service
    host_name          DC,SDC
    service_description C:\ Drive Space
    check_command       check_nt!USEDISKSPACE!-l c -w 80 -c 90
}

# Create a service for monitoring the W3SVC service
# Change the host_name to match the name of the host you defined above

define service {
    use                generic-service
    host_name          DC,SDC
    service_description W3SVC
    check_command       check_nt!SERVICESTATE!-d SHOWALL -l W3SVC
}

```

Figure 19 : Modification du define service dans windows.cfg

14. Supervision des serveurs Linux

Edition du fichier localhost.cfg pour ajouter les hotes Nagios et Zimbra. Modification du hostgroup et des services associes.

```
sudo nano /usr/local/nagios/etc/objects/localhost.cfg
```

```
GNU nano 6.2 /usr/local/nagios/etc/objects/localhost.cfg *
#####
# LOCALHOST.CFG - SAMPLE OBJECT CONFIG FILE FOR MONITORING THIS MACHINE
#
#
# NOTE: This config file is intended to serve as an *extremely* simple
#       example of how you can create configuration entries to monitor
#       the local (Linux) machine.
#
#####

#####
#
# HOST DEFINITION
#
#####

# Define a host for the local machine

define host{
    use             linux-server
    host_name       Nagios
    alias           Serveur de supervision
    address         172.20.0.31
}

define host{
    use             linux-server
    host_name       Zimbra
    alias           Serveur de messagerie
    address         172.20.0.15
}
█
```

Figure 20 : Modification du define host dans localhost.cfg

```

GNU nano 6.2 /usr/local/nagios/etc/objects/localhost.cfg *
# Define a service to check the swap usage the local machine.
# Critical if less than 10% of swap is free, warning if less than 20% is free

define service {
    use                local-service          ; Name of service template to use
    host_name          Nagios,Zimbra
    service_description Swap Usage
    check_command       check_local_swap!20%!10%
}

# Define a service to check SSH on the local machine.
# Disable notifications for this service by default, as not all users may have SSH en>

define service {
    use                local-service          ; Name of service template to use
    host_name          Nagios,Zimbra
    service_description SSH
    check_command       check_ssh
    notifications_enabled 0
}

# Define a service to check HTTP on the local machine.
# Disable notifications for this service by default, as not all users may have HTTP e>

define service {
    use                local-service          ; Name of service template to use
    host_name          Nagios,Zimbra
    service_description HTTP
    check_command       check_http
    notifications_enabled 0
}

```

Figure 21 : Modification du hostgroup : ajout des serveurs Nagios et Zimbra

15. Verification dans l'interface web

Verification de l'ajout des machines dans les menus Hosts et Map de l'interface web Nagios.

Current Network Status

Last Updated: Tue Mar 17 15:17:47 UTC 2026
 Updated every 90 seconds
 Nagios® Core™ 4.4.8 - www.nagios.org
 Logged in as *nagiosadmin*

[View Service Status Detail For All Host Groups](#)
[View Status Overview For All Host Groups](#)
[View Status Summary For All Host Groups](#)
[View Status Grid For All Host Groups](#)

Host Status Totals

Up	Down	Unreachable	Pending
1	1	0	4
All Problems		All Types	
1		6	

Service Status

Ok	Warning	Unknown
0	0	0
All Problems		
0		

Host Status Details For All Host Groups

Limit Results: ▾

Host	Status	Last Check	Duration	Status Information
DC	DOWN	03-17-2026 15:16:40	0d 0h 1m 7s	CRITICAL - Host Unreachable (172.20.0.10)
Nagios	UP	03-17-2026 15:17:30	0d 0h 1m 7s+	PING OK - Paquets perdus = 0%, RTA = 0.11 ms
SDC	PENDING	N/A	0d 0h 1m 7s+	Host check scheduled for Tue Mar 17 15:18:20 UTC 2026
Zimbra	PENDING	N/A	0d 0h 1m 7s+	Host check scheduled for Tue Mar 17 15:19:10 UTC 2026
hplj2605dn	PENDING	N/A	0d 0h 1m 7s+	Host check scheduled for Tue Mar 17 15:20:00 UTC 2026
linksys-srw224p	PENDING	N/A	0d 0h 1m 7s+	Host check scheduled for Tue Mar 17 15:20:50 UTC 2026

Results 1 - 6 of 6 Matching Hosts

Figure 22 : Vue du menu Hosts dans l'interface web

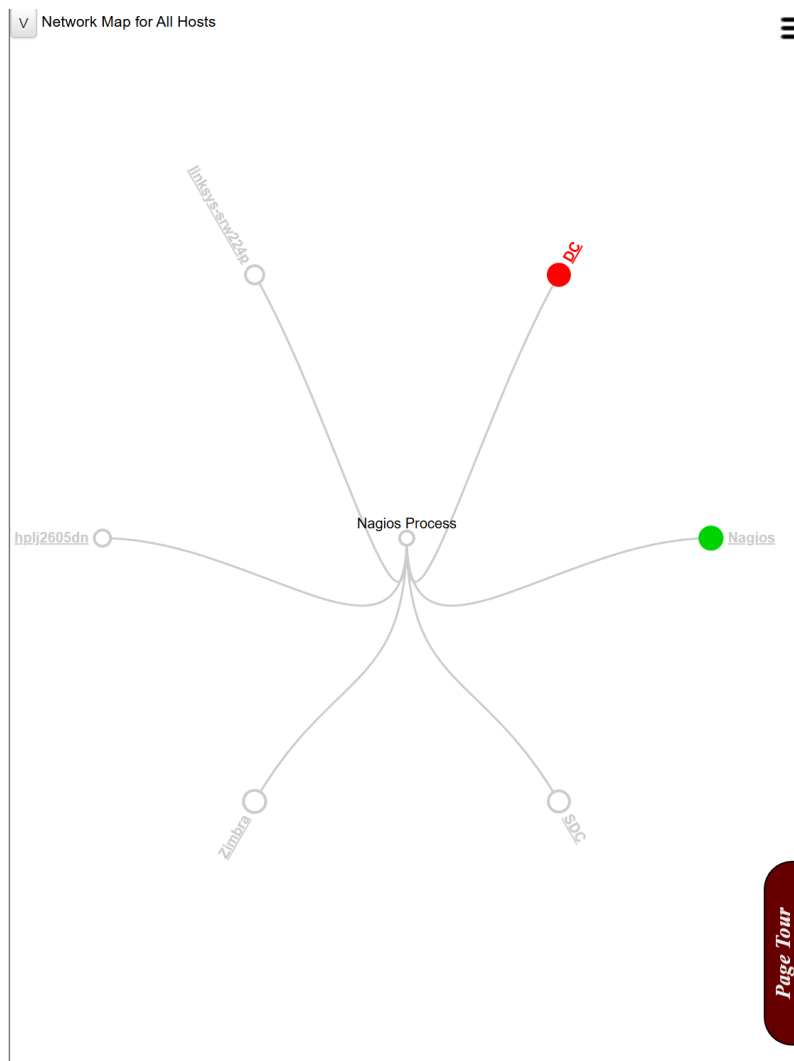


Figure 23 : Vue de la Map dans l'interface web Nagios

16. Configuration de la notification par mail

Configuration de ssmtp, du fichier revaliases, de resolv.conf et de contacts.cfg afin d'activer l'envoi de notifications par mail depuis Nagios.

```
sudo nano /etc/ssmtp/ssmtp.conf
sudo nano /etc/ssmtp/revaliases
sudo nano /etc/resolv.conf
sudo nano /usr/local/nagios/etc/objects/contacts.cfg
```

```
GNU nano 6.2 /etc/ssmtp/ssmtp.conf *
#
# Config file for sSMTP sendmail
#
# The person who gets all mail for userids < 1000
# Make this empty to disable rewriting.
root=postmaster

# The place where the mail goes. The actual machine name is required no
# MX records are consulted. Commonly mailhosts are named mail.domain.com
mailhub=mail.stadiumcompany.com

# Where will the mail seem to come from?
#rewriteDomain=

# The full hostname
hostname=nagios.m2l.local

# Are users allowed to set their own From: address?
# YES - Allow the user to specify their own From: address
# NO - Use the system generated From: address
#FromLineOverride=YES
```

Figure 24 : Configuration du fichier contacts.cfg avec le mail Stadium Company

```
GNU nano 6.2 /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.
nameserver 172.20.0.10
nameserver 1.1.1.1
search stadiumcompany.com
```

Figure 25 : Configuration du fichier resolv.conf